



Spot an Error

And WIN Award!

You can win one thousand rupees plus a certificate from the editor of Electronics For You magazine if you spot and report any factual error in the latest issue of the magazine. Your report will be published in a coming issue of the magazine, giving credit to you.

Though every care is taken in preparation of the content for the magazine, an error can slip in at times. So we invite our learned readers to point out any error that they spot, for the benefit of the other readers.

Rules:

- The error has to be reported before end of the month that is mentioned on cover of the magazine
- The report has to be sent in writing or by email addressed to the Editor, EFY Magazine (editsec@efyindia.com)
- The error should be of factual nature, in the information published in the magazine, and not a grammatical or syntax error
- The correct information should also be provided along with the error spotted by you
- In case more than one person reports the same error, the one whose report reaches us the earliest will get the award money
- In case two or more persons jointly report an error, the award money will be divided amongst them
- EFY employees, their family members and friends, and EFY associates and their family members and friends are not eligible
- Decision of the editor will be final in this matter

electronics
FOR YOU

Networks (NSN) have already begun discussing implementation of projects such as smart delivery and smart retailing, by connecting the next billion devices to the Internet and using that network to provide value-added solutions. The technologies used here are those like UMTS.

3G/UMTS, in its initial phase, offers theoretical bit rates of up to 384 kbps in high mobility situations such as urban and suburban areas, 144 kbps in rural areas and 2 Mbps in stationary or nomadic user environments for short-range applications.

"Though the world is moving towards 3G and LTE, 2G is still relevant in low-data-rate M2M applications. It will take a long time for 3G/LTE prices (for hardware) to match current 2G module price levels of US\$ 7 to 10," says Gopakumar, director sales, Fides Electronics Pvt Ltd.

UMTS requires new base stations and new frequency allocations. Not all countries have the same UMTS-2100 standard bands. Very high speeds at around 21 Mbit/s are possible on this network by using high-speed download packet access (HSDPA), and are recommended for content consumption devices. Long Term Evolution (LTE) will further increase this to 100 Mbit/s downlink and 50 Mbit/s uplink.

So will this mean more expense in test equipment? "Today's T&M are ready for LTE-Advanced applications. The same spectrum analyser can be used for 3G spectrum measurement application and for LTE spectrum measurement as well. Moreover, Universal Wireless Test Set is available which makes all 3G LTE measurements along with IEEE and Bluetooth measurements in single box," explains Madhukar Tripathi, regional manager, Anritsu India Pvt Ltd.

The best fit. Video telephony, and enabling similar streaming content services over a very large area, is the main application that UMTS can serve. Along with HSDPA or LTE, the high data throughput enables multi-user video conferences and the ability to

execute online classes. It also enables location calculation based solely on the cell coverage, thus removing the requirement for any additional hardware. Assisted-GPS functionality is vastly improved here.

Tunnelled Direct Link Setup (TDLS)

Tunnelled Direct Link Setup is an implementing technology which enables devices to connect directly to one another after they have joined a traditional Wi-Fi network. The establishment of direct links between TDLS-certified devices requires no user intervention.

The TDLS discovery process begins when one TDLS device sends a TDLS discovery request to another device, via the AP or GO (group owner). If the target device is also TDLS compliant, then it sends a TDLS discovery response directly to the initiator, providing information on its capabilities, including supported rates and channels. In addition to providing the information, the TDLS discovery frame exchange may be used to measure the relative signal strengths of the AP and candidate TDLS device. This would inform a decision on whether a direct link would be beneficial. TDLS set up can also be done without TDLS discovery.

The best fit. "Some of the use cases best suited to the TDLS network are streaming media to a Wi-Fi TV, transfer of files from a PC to a smartphone, streaming audio to a Wi-Fi speaker, and wireless printing using direct link," explains Lavanya Sundarajan, project lead, PES, Mistral Solutions. It is also a great solution for wireless data backup such as when using the Time Machine backup solution.

In the home and enterprise environments, TDLS technology greatly enhances the efficiency of communication between different devices in a network, such as streaming (video and audio), data migration and data backup. ●

The author is a tech correspondent at EFY Bengaluru